

From a single characterisation snapshot to end-of-life: a PyBaMM-trained neural operator for second-life LFP RUL

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Second-life reuse of used lithium-iron-phosphate (LFP) cells for stationary energy storage is bottlenecked by characterisation cost. Per-cell PyBaMM [1] calibration of degradation parameters against measured state-of-health (SoH) trajectories requires at least 400 cycles per cell [2] to reliably extrapolate to end-of-life; below that, the fitted rates become under-determined. Prior second-life work [3] and universal-differential-equation degradation [4] improve the quality of the per-cell fit but do not shrink the cycling budget itself. Reducing that budget requires transferring knowledge from a synthetic parameter-space corpus to individual cells rather than fitting each from scratch.

The approach has two stages. A corpus of 300 synthetic degradation trajectories is first generated with PyBaMM's DFN model using the Prada2013 [2] LFP chemistry and per-cell beginning-of-life parameters identified from OCV, GITT, HPPC and RPT tests on a fresh supplier-B cohort. A Sobol sweep across five degradation channels — SEI kinetics, plating current, positive- and negative-electrode active-material loss, and SEI molar volume — spans the physically-plausible parameter space at 25 °C, producing 300 trajectories paired with the degradation-parameter vector θ that generated each. A DeepONet [5] variant is then trained to map a five-stream fingerprint (DCIR, RPT features, the first- K measured SoH values, θ , and cycling protocol) onto the SoH trajectory at any queried cycle. Conditioning on θ lets the operator absorb electrochemistry directly from the identification step rather than reconstructing it from characterisation alone.

End-to-end validation on a single supplier-B cell drives both the BOL identification and the degradation-parameter fit from that cell's own data. Fitting the four SEI and plating parameters against its 150-cycle measured Longterm record yields **0.3 pp SoH RMSE**; the resulting PyBaMM long-run reaches first-life end-of-life (SoH = 0.80) at **cycle 3921** with a smooth monotonic fade, consistent with major LFP manufacturer datasheet cycle-life claims at 0.5 C. In this presentation the trained operator's predictions on 20 held-out cells across three commercial LFP suppliers will be shown, together with the identifiability limits of parameter fitting from short cycling records and extensions to non-isothermal deployment.

References

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